COORDINATE GEOMETRY

Distance & Midpoint Formula

Grade 6 Mathematics

1. What Is the Coordinate Plane?

The coordinate plane is a flat surface made up of two number lines that cross each other at right angles.

The **horizontal line** is called the **X-axis** — it runs left and right. The **vertical line** is called the **Y-axis** — it runs up and down. Where they meet is called the **origin (0, 0)**.

Together, these axes create a grid where we can locate any point using two numbers!

■ Think of it as a playground for points!

2. Understanding Coordinates

Coordinates help us locate exact positions on the coordinate plane. Every point is written as an **ordered pair (x, y)**, where:

Symbol	Meaning	Direction
x	Horizontal position	Move left or right first
y	Vertical position	Move up or down second

3. Distance Between Two Points

In coordinate geometry, **distance** tells us how far apart two points are on the coordinate plane — the length of the straight line connecting them.

The Distance Formula

Distance = √[(x■ – x■)² + (y■ – y■)²]

Step 1: Subtract the x-coordinates: $(x_2 - x_1)$

Step 2: Subtract the y-coordinates: $(y_2 - y_1)$

Step 3: Square each difference to remove negatives

Step 4: Add the two squared values together

Step 5: Take the square root to find the actual distance

Worked Example

Find the distance between A(2, 3) and B(7, 8).

```
x■ = 2, y■ = 3 x■ = 7, y■ = 8
Distance = √[ (7–2)² + (8–3)² ]
= √[ 5² + 5² ]
= √[ 25 + 25 ]
= √50 ≈ 7.07 units
```

✔ The distance between (2, 3) and (7, 8) is √50 ≈ 7.07 units

Practice

■ Find the distance between the points (1, 2) and (5, 6). Use the formula and show each step.

4. The Midpoint

The **midpoint** is the exact middle point between two points on a coordinate plane — think of it as the *halfway spot*!

To find the midpoint, take the average of the x-coordinates and the average of the y-coordinates.

The Midpoint Formula

Midpoint = ((x■ + x■) / 2 , (y■ + y■) / 2)

Step 1: Add the two x-coordinates and divide by 2 → gives the middle x value

Step 2: Add the two y-coordinates and divide by 2 → gives the middle y value

Step 3: Write the result as an ordered pair (x, y)

Worked Example

Find the midpoint between A(4, 6) and B(10, 2).

```
Midpoint = ( (4 + 10) / 2 , (6 + 2) / 2 )
= ( 14 / 2 , 8 / 2 )
= ( 7 , 4 )
```

■ The midpoint between (4, 6) and (10, 2) is (7, 4)

Practice

■ Find the midpoint between the points (3, 5) and (7, 9). Remember the formula and show your working.

5. Distance vs Midpoint: Quick Comparison

	Distance	Midpoint
What it finds	How far apart two points are	The exact centre point
Formula uses	Subtraction, squaring, square root	Addition, division by 2
Result type	A single number (length)	An ordered pair (x, y)
Key question	How far?	Where is the middle?

6. Common Mistakes to Avoid

■ **Forgetting to square** — Always square each difference in the distance formula *before* adding them.

■ **Skipping the division by 2** — In the midpoint formula you must divide *each* coordinate sum by 2.

■ **Mixing up x and y** — Substitute x-values in the x position and y-values in the y position.

■ **Arithmetic slips** — Double-check every subtraction, addition and square-root step.

7. Real-Life Uses

- ■ **Maps & GPS:** The distance formula calculates how far apart two locations are.
- ■ **Meeting Point:** The midpoint formula finds the fair halfway spot between two friends.
- ■ **Urban Planning:** Both formulas help divide land, design parks, and place landmarks.

8. Challenge Question!

Use **both** formulas to answer this question:

Find the **distance** AND the **midpoint** between the points (1, 4) and (5, 10).

Distance =	
Midpoint =	

Summary: What We Learned Today

- ■ The coordinate plane uses X and Y axes to locate any point as (x, y).
- ■ The **distance formula** √[(x■–x■)²+(y■–y■)²] measures how far apart two points are.
- ■ The **midpoint formula** ((x■+x■)/2, (y■+y■)/2) finds the centre between two points.
- ■ Distance gives a *number*; midpoint gives an *ordered pair*.
- ■ Avoid common mistakes: always square, always divide by 2, and double-check arithmetic!

■ Keep practicing, stay curious — coordinate geometry is your superpower for solving real-world problems!